

Brain Tumor Segmentation Using Neural Ordinary Differential Equations with UNet-Context Encoding Network

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Abstract

Glioblastoma Multiforme (GBM) are the most aggressive brain tumor types and because of their heterogeneity in shape and appearance, their segmentation becomes a challenging task. Automated brain tumor segmentation using Magnetic Resonance Imaging (MRI) plays a key role in disease diagnosis, surgical planning, and brain tumor tracking. Medical image segmentation using deep learning-based U-Net architectures are the state-of-the-art. Despite their improved performance, these architectures require optimization for each segmentation task. Introducing a continuous depth learning with context encoding in deep CNN models for semantic segmentation enable 3D image analysis quantifications in many applications. In this work, we propose Neural Ordinary Differential Equations (NODE) with 3D UNet-Context Encoding (UNCE), a continuous depth deep learning network for improved brain tumor segmentation. We showed that these NODEs can be implemented within the U-Net framework to improve segmentation performance. This year we participated for the Brain Tumor Segmentation (BraTS) continuous evaluation and our model was trained using the same MRI image sets of RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021. Our model is evaluated on unseen hold-out data included i) the BraTS 2021 Challenge test data, ii) SSA adult patient populations of brain diffuse glioma (Africa-BraTS), and iii) from another independent pediatric population of diffuse intrinsic pontine glioma (DIPG) patients. The mean DSC for the

BraTS test dataset are: 0.797797 (ET), 0.825647 (TC) and 0.894891 (WT) respectively. For the Africa-BraTS dataset the performance of our model improves which indicating the generalizability of our model to new, out-of-sample populations for adult brain tumor cases.

Keywords Glioblastoma – Brain Tumor Segmentation – Neural Ordinary Differential Equations – Deep neural network – U-Net

1 Introduction

High-grade gliomas with grade III and IV according to World Health Organization (WHO) are the most aggressive and severe malignant brain tumors of the central nervous system (CNS) in adults, with extreme intrinsic heterogeneity in appearance, shape, and histology [1]. Patients diagnosed with the most aggressive type of brain tumor have a median survival time of two years or less [2]. The precise and robust segmentation of brain tumors and the prediction of the patient's overall survival are the basis for subsequent clinical steps and supportive for improving the treatment possibilities of the disease [3]. Conventional segmentation models based on clinical information and handcrafted features from multi-modal (MRI) neuroimaging are time-consuming, may be inaccurate due to human error, and also require anatomical knowledge. The manual extraction of handcrafted features is based on the prior knowledge of disease and can also be limited to the existing imaging techniques. This often limits the ability to take full advantage of all rich information embedded in the multi-modal (MRI) neuroimages for the brain tumor segmentation [4, 5].

Automatic brain tumor segmentation is a difficult task. Compared to the classical hand-crafted feature extraction methods, deep learning models can learn a hierarchy of features that could represent the rich embedded information as hierarchical abstraction at many different levels [6–9]. In addition, the robustness in performance remains an open and difficult challenge in deep learning based models. Recent studies have shown that deep learning tools for brain tumor segmentation have been developed and achieved a better performance compared to classical models [3, 10–13]. The current state-of-the-art for the medical image segmentation models are deep learning-based networks that follow U-Net architecture [14]. These models have made it feasible to build large-scale trainable models with a large number of receptive field at high computational cost of millions of parameters. To achieve reasonable performance, these models often require adding layers to increase the receptive field which comes at the cost of more parameters, computation, and memory requirements. It is a very challenging task to optimize deep learning models for every segmentation task. Therefore, proper regularization and hyper-parameter tuning are required for developing an efficient deep learning network.

In the previous BraTS challenge we implemented context encoding [15] with deep CNN models which have shown promise for semantic segmentation of brain tumors due to improved representative feature learning. We propose to use a continuous depth deep learning model called Neural Ordinary Differential Equations (NODEs) [16, 17] within UNet-Context Encoding (UNCE) [18] framework for automatic brain tumor segmentation. We implemented this technique on the BraTS 2022 challenge datasets. Our model performance indicates that these NODEs can be used within U-Net framework to improve the segmentation results.

2 Method

2.1 Datasets

For training and validation data for the BraTS 2022 challenge, we use the same datasets provided by the RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021 dataset obtained from multiple different institutions under standard clinical conditions [1]. In the BraTS Challenge 2022 Task, a total of

1251 cases with voxel-level annotations of different tumor sub-tissues: necrotic (NC), peritumoral edema (ED), and enhancing tumor (ET) are presented.

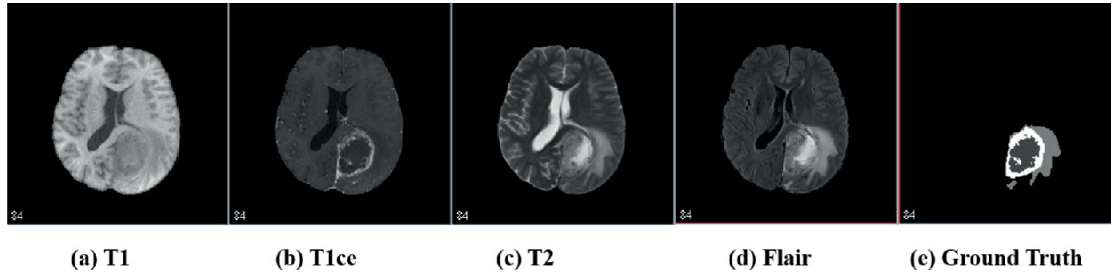


Fig. 1. Examples of four different MRI modalities: (a) T1, (b) T2, (c) T1ce, (d) FLAIR of a training sample with Ground Truth.

For online evaluation in the validation phase, there are 219 cases provided without any associated ground truth. Ground truth annotations of every tumor sub-region for brain tumor segmentation were approved by expert neuroradiologists [19–22]. Each patient case has four MRI modalities: T1-weighted (T1), T1-weighted contrast enhancement (T1ce), T2-weighted (T2), and T2-weighted fluid-attenuated inversion recovery (T2-FLAIR). All modality sequences are co-registered, skull-stripped, denoised, and bias-corrected [23]. Figure 1 shows an example from training cases of four MRI modalities (T1, T1ce, T2 and FLAIR).

2.2 UNCE-NODE Network

Neural-ODEs: Approximation of the derivative of any differentiable function using neural network provides more flexibility in the data modeling process. Residual neural networks appear to follow the modeling pattern of an ODE: namely that the continuous relationship is modeled at the level of the derivative. This is the main idea of neural ODEs: a chain of residual blocks in a neural network is basically a solution of the ODE with the Euler method. Euler’s method is a discretization of the continuous relationship between the input and output domains of the data. Neural networks are also discretization of this continuous relationship, only the discretization is through hidden states in a latent space. Residual neural networks create a pathway through this latent space by allowing states to depend directly on each other, just like the updates in Euler’s method.

Considering a neural network where the hidden states all have the same dimension. Each hidden state depends on a neural layer, f , which itself depends on parameters θ_t , where t is the layer depth. Then

$$h_{t+1} = f(h_t, \theta_t) \quad (1)$$

If we have residual network, then this looks

$$h_{t+1} = h_t + f(h_t, \theta_t) \quad (2)$$

where $t \in \{0 \dots T\}$, $h_t \in \mathbb{R}^d$, and f a differentiable function. In ResNets, f consists of several convolutional layers. The update with residual $f(h_t)$ can be seen as a $\Delta t = 1$ step of an Euler discretization of a continuous transformation. When we let $\Delta t \rightarrow 0$ we take more, smaller steps using more layers, which in the limit becomes an ordinary differential equation (ODE), specified by a neural network:

$$\lim_{\Delta t \rightarrow 0} \frac{h_{t+\Delta t} - h_t}{\Delta t} = \frac{\delta h_t}{\delta t} = f(h(t), t, \theta) \quad (3)$$

where $h(t)$ is the value of the hidden state evaluated for some time t , which we understand as a continuous parametrization of layer depth. We consider the continuous limit of each discrete layer in the network. Instead of a discrete number of layers between the input and output domains, we allow the progression of the hidden states to become continuous.

A neural network is a differentiable function, so we can train it with gradient-based optimization routines. Ordinary Differential Equations (ODE) can be solved using standard ODE solvers such as Runge-Kutta [24,

25]. Since the model is a learnable ODE, we use an ODE solver to evolve the input to an output in the forward pass and calculate a loss. In particular, for weight updating during back propagation, we need a gradient of loss function by the input and the dynamics parameters. Unfortunately, during the forward pass an ODE solver might require hundreds of function evaluations, which may lead to exploding memory requirements. The mathematical trick to retrieve the gradient of the Neural ODE loss with respect to the network parameters by solving another ODE backwards in time, sidestepping the need to store intermediate function evaluations, is called adjoint sensitivity method (for more details see Algorithm 1 in Chen et al., 2018 [17]).

The advantages and motivation to use NODE networks for semantic segmentation are: They are memory efficient given intermediate computations (e.g. activation maps), so we don't need to store all the parameters and gradients while back propagating. Adaptive computation: we can balance speed and accuracy with the discretization scheme, moreover, having it different while training and inference. Parameter efficiency: the parameters of nearby "layers" are automatically tied together.

In this work, we have implemented a continuous depth deep neural network within an UNCE network framework that integrates multiple volumetric MRI processing tasks. Inspired by the work of context encoding networks [13], the UNCE architecture is substantially augmented for brain tumor segmentation.

Two models (a) UNCE deep learning baseline and (b) UNCE deep learning with NODE for tumor segmentation are trained using the BraTS 2021 challenge datasets. The overall pipeline for the brain tumor segmentation using UNCE-NODE is shown in Fig. 2. The UNCE captures global texture features using a semantic loss to provide regularization in training. The architecture consists of encoding, context encoding, and decoding modules. The encoding module extracts high-dimensional features of the input. The context encoding module produces updated features and a semantic loss to regularize the model by ensuring all segmentation classes are represented. The decoding module reconstructs the feature maps to produce segmentation masks as output. UNCE-Neural ODEs present a new architecture with much potential for reducing parameter and memory costs, improving the performance. Figure 2 below shows the architecture of the 3D UNCE-baseline and UNCE-NODE models and its different components. The U-Net architecture consists of a contracting and an expanding path that aims to build a bottleneck in its innermost part through a combination of convolution, instance norm, and softmax operations. After this bottleneck, the image is reconstructed by combining convolutions and upsampling.

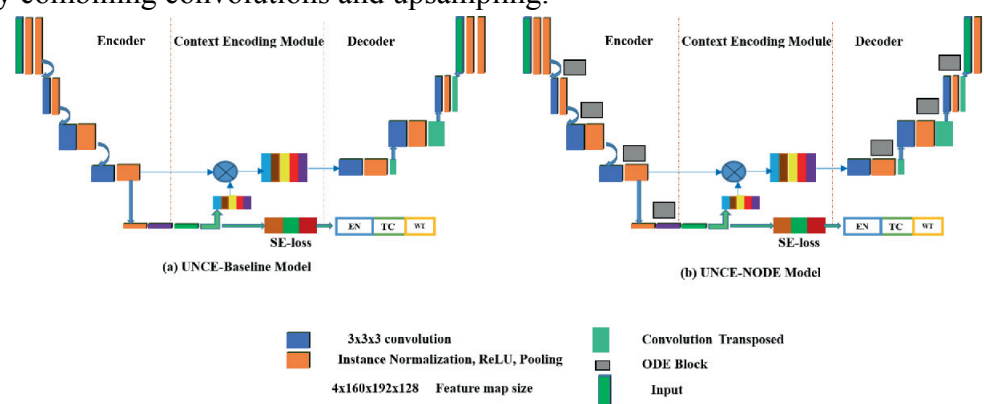


Fig. 2. The Brain Tumor Segmentation pipeline. (a) UNCE-Baseline and (b) UNCE-NODE are fully automatic, end-to-end deep learning pipelines that segment brain tumor sub-regions from 3D MRI across different modalities.

Implementation: To justify the heterogeneity of the medical field, all images are resampled to the same target spacing. The MRI scans provided for the competition are co-registered, skull-stripped, denoised and bias corrected. Since the dataset was collected from different institutions and MRI scanners, the intensities show substantial variations across examples. Consequently, we perform normalization of all examples to

have zero mean and unit standard deviation. The dimension of each training sample is $240 \times 240 \times 155$. The size of the images is reduced by cropping to a size of $192 \times 160 \times 128$ to manage computational memory and cost. To generate additional training images, we augment data by adding uniform white noise with limited amplitude. A critical feature of the proposed UNCE is the context encoding module, which computes scaling factors related to the representation of all classes. These factors are learned simultaneously in the training phase via the semantic loss error regularization, defined by L_{se} . The scaling factors capture global information of all classes, essentially learning to mitigate the training bias that may arise due to imbalanced class representation in data. To calculate the Semantic Error loss (SE-loss), we construct another fully connected layer with a sigmoid activation function upon the encoding layer, to predict object classification in the image [13]. Accordingly, the final loss function consists of 2 terms:

$$L = L_{dice} + L_{se} \quad (4)$$

where L_{dice} is a Dice calculated by the difference between prediction and ground truth, and L_{se} is the semantic loss. Dice loss is computed as:

$$L_{dice} = 1 - DSC \quad (5)$$

where DSC is dice similarity coefficient [26]. The DSC is defined as,

$$DSC = \frac{2TP}{FP + 2TP + FN} \quad (6)$$

where TP, FP and FN are the numbers of true positive, false positive and false negative, respectively. We use an Adam optimizer [27] with initial learning rate of $l_{r0} = 0.0001$ in training phase, and the learning rate (l_{ri}) is gradually reduced by following:

$$l_{ri} = l_{r0} \star \left(1 - \frac{i}{N}\right)^{0.9} \quad (7)$$

where i is epoch counter, and N is a total number of epochs in training. We used ODE solvers from the torchdiffeq python package [17] and used the fifth-order “dopri5” solver, with a 10^{-3} tolerance.

Table 1. Network configurations for UNCE for the BraTS 2022 Challenge Dataset

Parameters	UNCE-Baseline	UNCE-NODE
Target spacing (mm):	$1 \times 1 \times 1$	$1 \times 1 \times 1$
Median image shape: at target spacing	$240 \times 240 \times 155$	$240 \times 240 \times 155$
Patch size:	$192 \times 160 \times 128$	$192 \times 160 \times 128$
Batch size:	2	2
Downsampling strides:	[2, 2, 2]	[2, 2, 2]
Convolution kernel sizes:	[3, 3, 3]	[3, 3, 3]
ODE Block:	No	Yes

Table 1 represents the variants of UNCE network configuration for the Brain Tumor Segmentation (BraTS) 2022 challenge. In the following sections, we summarized the detailed information for the implementation of UNCE network.

We implement the context-aware deep learning network in PyTorch and train the network on an NVIDIA V100 HPC platform using the BraTS training dataset (1251 cases). To train the network, we randomly split the dataset with 80% of data for training and the remaining 20% of the data for validating the trained model. All configurations of UNCE are trained for 300 epochs and the best performing versions based on the validation set are retained for testing. Effective training of the network is observed by the monotonically

decreasing loss and the corresponding increase in training dice score. Images are predicted by using the best performing version of the trained model. Once the network is fully trained, the performance of the network is evaluated using the BraTS 2022 validation dataset (219 cases) utilizing the online submission process made available by the challenge organizers. Figure 3 presents an example segmented tumor and the corresponding ground truth of T2 images.

3 Results

The 1251 cases for the brain tumor segmentation task are used for training each configuration of UNCE models. We used the best performing UNCE models using the training dataset provided by the Brain Tumor Segmentation (BraTS) Challenge 2022 organizer to predict the validation cohort.

Table 2. Online Evaluation on Validation dataset(219 cases)

Model	Statistical parameter	Dice Score			Hausdorff95		
		ET	TC	WT	ET	TC	WT
UNCE-Baseline	Mean	0.7810	0.8244	0.9095	15.2337	12.6095	4.5294
	std	0.2389	0.2431	0.0846	65.2718	50.1448	7.6226
	Median	0.8590	0.9255	0.9382	2.2360	2.2360	2.8284
	25quantile	0.7844	0.8300	0.8939	1.4142	1.4142	1.7320
	75quantile	0.9055	0.9554	0.9534	3.09331	5.9154	4.2426
UNCE-NODE	Mean	0.8008	0.8500	0.9108	19.6636	6.9889	4.2722
	std	0.2253	0.2006	0.0748	77.3861	26.3836	5.9913
	Median	0.8644	0.9283	0.9328	2	2.2360	2.4494
	25quantile	0.7983	0.8506	0.8920	1.4142	1.4142	1.7320
	75quantile	0.9091	0.9571	0.9520	3	5	4.3155

Table 3. Online Evaluation on Testing dataset

Test Dataset	Dice Score			Hausdorff95		
	ET	TC	WT	ET	TC	WT
BraTS 2021	0.797797	0.825647	0.894891	23.70354	20.33859	6.05848
Africa-BraTS	0.817089	0.904995	0.944727	4.685421	3.460575	4.27374
DIPG	–	0.365294	0.835283	–	145.4344	9.215518

We generated the segmented mask from different variants of UNCE configurations and submitted for the online evaluation using the validation datasets. The online evaluation results with DSC and HD95 are reported in Table 2. The dice scores of the UNCE-NODE model are 0.8008, 0.8500, and 0.9108 for enhancing tumor (ET), tumor core (TC), and whole tumor (WT), respectively. The Hausdorff95 distances of the same model are 19.66 for ET, 6.98 for TC, and 4.27 for WT. The evaluation results show a further improvement of dice scores compared to the UNCE-baseline model.

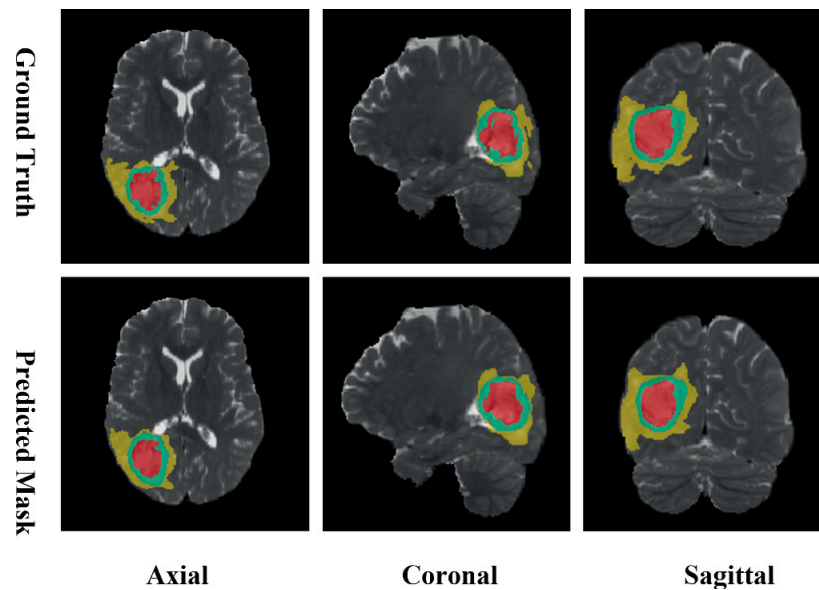


Fig. 3. T2 image overlaid with ground truth and predicted mask. From left to right: axial, coronal, and sagittal view respectively

In the testing phase, the unseen hold-out testing data included the BraTS 2021 Challenge test data, as well as new data from out-of-sample sources including i) an independent multi-institutional dataset covering underrepresented SSA adult patient populations of brain diffuse glioma (Africa-BraTS), and ii) from another independent pediatric population of diffuse intrinsic pontine glioma (DIPG) patients. According to the requirement of the challenge, we prepared a docker image of our model and our submitted docker image has been segmented all the unseen hold-out test examples. We have submitted our final model to evaluate the segmentation of the test dataset. The performance of mean dice score coefficient (DSC), and Hausdorff distance are shown in Table 3. The online evaluation shows that the average DSC of WT, ET and TC in the testing phase is closer to that of in the validation phase for the BraTS testing cohort. For the Africa-BraTS dataset the performance of our model improves. However, the dice score of the tumor core (TC) is very low for the DIPG dataset, which we believe there is domain shift in the data distribution between the source (BraTS) and target dataset (DIPG).

Figure 3 shows the T2 MRI image overlaid with ground truth in the first row and predicted segmentation mask in the second row respectively for the same case.

4 Discussion

This manuscript explains the application of NODE within the UNCE framework in the Multi-Modality Brain Tumor Segmentation (BraTS) Challenge 2022. The UNCE-NODE model is evaluated on the unseen hold-out testing data included the BraTS 2021 Challenge test data, as well as new data from out-of-sample sources including i) Africa-BraTS, and ii) DIPG patients. For the Africa-BraTS dataset the performance of our model improves which indicating the generalizability of our model to new, out-of-sample populations for adult brain tumor cases. However, the dice score of the tumor core (TC) is very low for the DIPG dataset, which we believe there is domain shift in the data distribution between the source (BraTS) and target dataset (DIPG). It would be interesting, if we could apply domain adaptation with transfer learning approach to perform domain-invariant segmentation. Transfer learning and domain adaptation are both approaches that aim to improve the generalizability of models to a target domain that differs from the source domain on which the

model was trained. In future work, we plan to investigate the generalizability of this approach to other age groups and data acquisition pipelines. We hope that this approach may be used to yield automated, objective assessments of age or domain varying patterns in other applications.

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